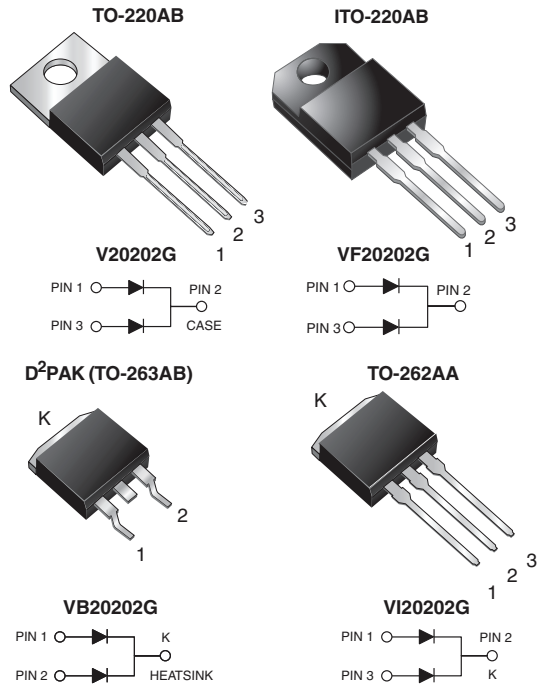
Dual High Voltage TMBS® (Trench MOS Barrier Schottky) Rectifier

Ultra Low V_F = 0.61 V at I_F = 5 A



FEATURES

- Trench MOS Schottky technology Gen 2
- Low forward voltage drop, low power losses
- High efficiency operation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 245 °C (for D²PAK (TO-263AB) package)
- Solder bath temperature 275 °C maximum, 10 s, per JESD 22-B106 (for TO-220AB, ITO-220AB, and TO-262AA package)
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

TYPICAL APPLICATIONS

For use in high frequency DC/DC converters, switching power supplies, freewheeling diodes, OR-ing diode, and reverse battery protection.

MECHANICAL DATA

Case: TO-220AB, ITO-220AB, D²PAK (TO-263AB), and TO-262AA

Molding compound meets UL 94 V-0 flammability rating Base P/N-M3 - halogen-free, RoHS-compliant, and commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102 M3 suffix meets JESD 201 class 1A whisker test

Polarity: as marked

Mounting Torque: 10 in-lbs max.

LINKS TO ADDITIONAL RESOURCES

3D Models

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	2 x 10 A
V_{RRM}	200 V
I_{FSM}	130 A
V_F at I_F = 10 A (T_A = 125 °C)	0.71 V
T_J max.	175 °C
Package	TO-220AB, ITO-220AB, D²PAK (TO-263AB), TO-262AA
Circuit configuration	Common cathode

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	V20202G	VF20202G	VB20202G	VI20202G	UNIT
Maximum repetitive peak reverse voltage		V_{RRM}	200				V
Maximum average forward rectified current (fig. 1)	per device	$I_{F(AV)}$	20				A
	per diode		10				
Maximum DC reverse voltage		V_{DC}	160				V
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load per diode		I_{FSM}	130				A
Voltage rate of change (rated V_R)		dV/dt	10 000				V/µs
Isolation voltage (ITO-220AB only) from terminal to heatsink, t = 1 min		V_{AC}	1500				V
Operating junction and storage temperature range		T_J, T_{STG}	-40 to +175				°C

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage per diode (1)	I_F = 5 A	T_A = 25 °C	V_F	0.76	-	V
	I_F = 10 A			0.84	0.92	
	I_F = 5 A	T_A = 125 °C		0.61	-	
	I_F = 10 A			0.71	0.8	
Reverse current per diode (2)	V_R = 160 V	T_A = 25 °C	I_R	0.3	-	µA
		T_A = 125 °C		0.5	-	mA
	V_R = 200 V	T_A = 25 °C		-	150	µA
		T_A = 125 °C		1.5	8	mA

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 5 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)							
PARAMETER		**SYMBOL**	**V20202G**	**VF20202G**	**VB20202G**	**VI20202G**	**UNIT**
Typical thermal resistance	per diode	$R_{\theta JC}$	2.8	5.0	2.8		°C/W
	per device	$R_{\theta JC}$	1.6	3.5	1.6		
	per device	$R_{\theta JA}$ (1)(2)	52	60	52		

Notes

(1) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$

(2) Free air, without heatsink

ORDERING INFORMATION (Example)					
PACKAGE	**PREFERRED P/N**	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
TO-220AB	V20202G-M3/4W	1.88	4W	50/tube	Tube
ITO-220AB	VF20202G-M3/4W	1.75	4W	50/tube	Tube
D^2PAK (TO-263AB)	VB20202G-M3/4W	1.39	4W	50/tube	Tube
D^2PAK (TO-263AB)	VB20202G-M3/8W	1.39	8W	800/reel	Tape and reel
TO-262AA	VI20202G-M3/4W	1.45	4W	50/tube	Tube

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

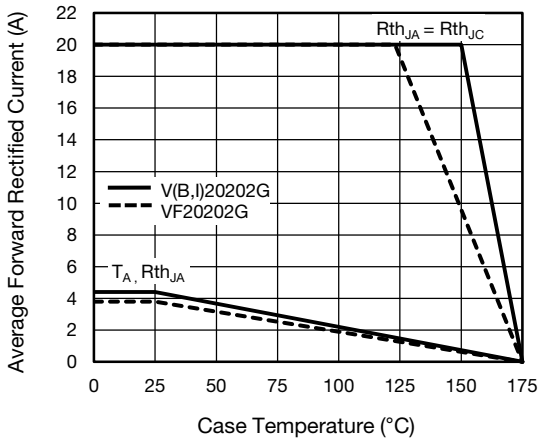


Fig. 1 - Maximum Forward Current Derating Curve
(D = Duty Cycle = 0.5)

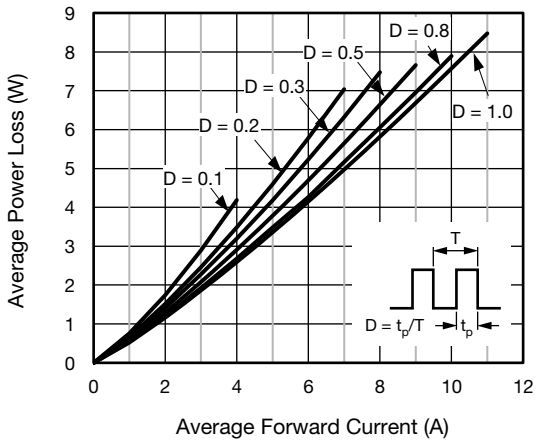


Fig. 2 - Forward Power Loss Characteristics Per Diode

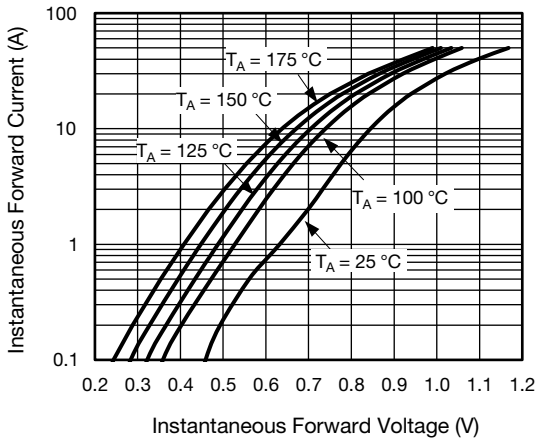


Fig. 3 - Typical Instantaneous Forward Characteristics Per Diode

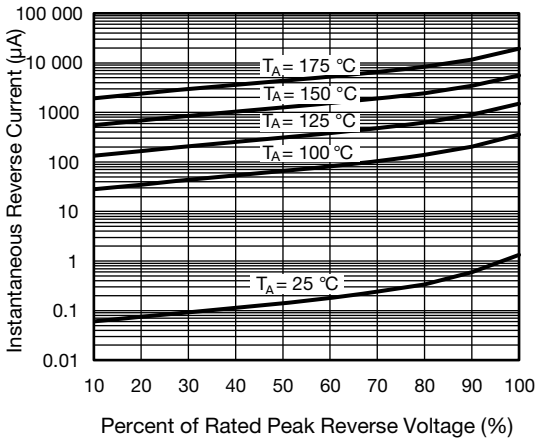


Fig. 4 - Typical Reverse Characteristics Per Diode

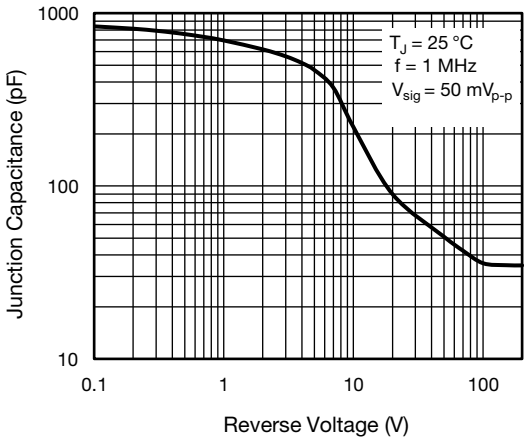


Fig. 5 - Typical Junction Capacitance Per Diode

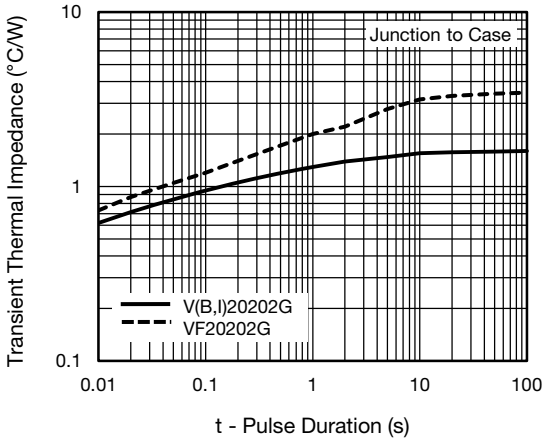
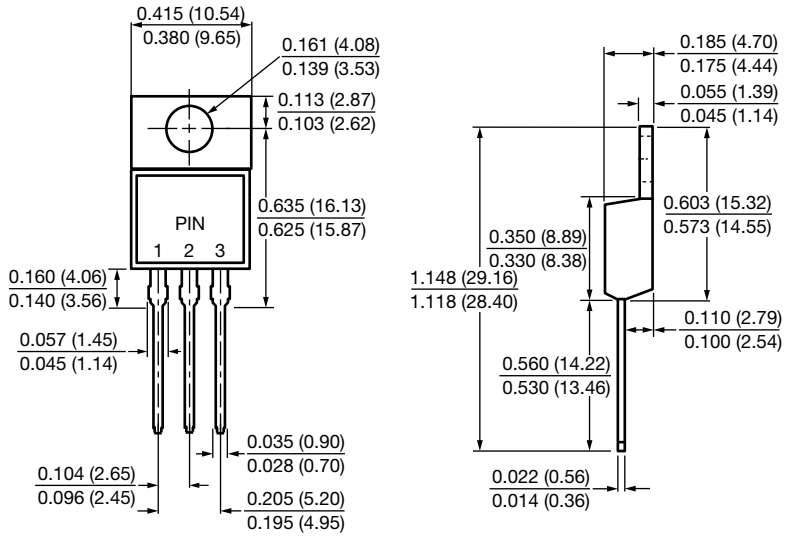


Fig. 6 - Typical Transient Thermal Impedance Per Device

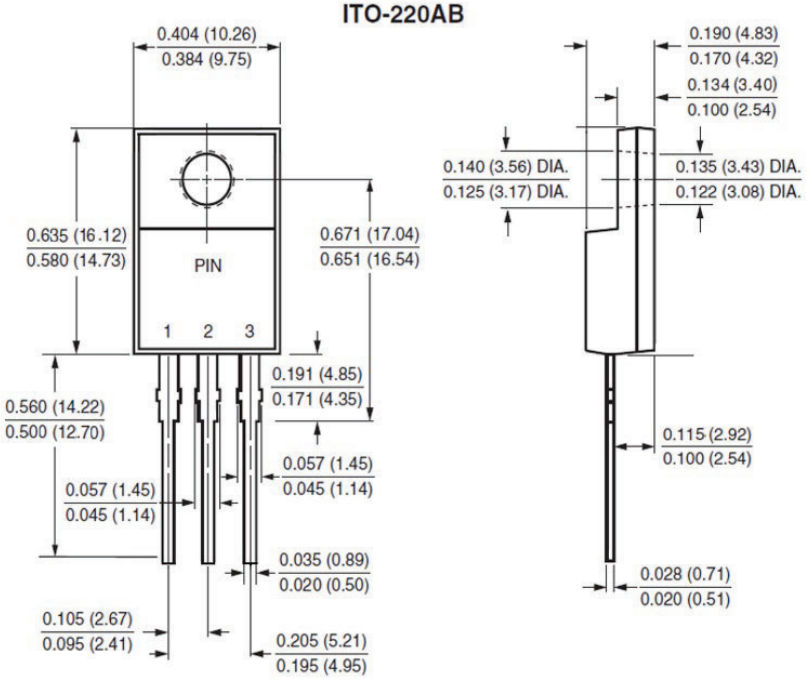
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

TO-220AB

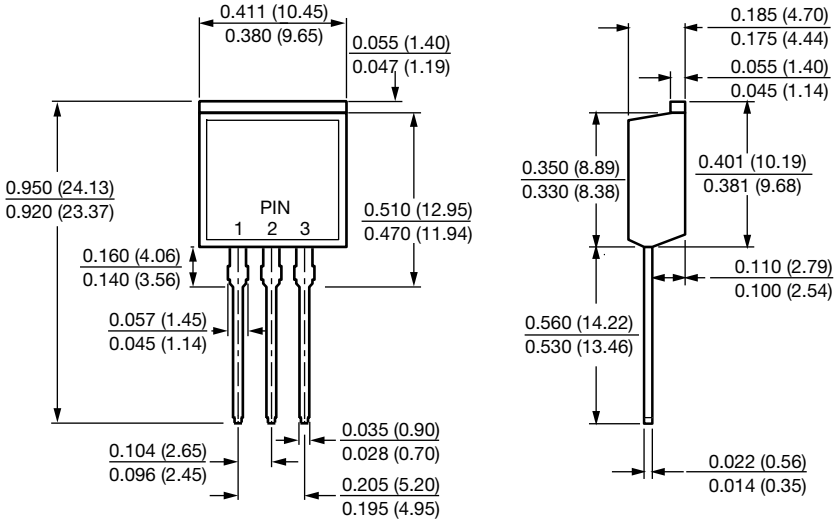


ITO-220AB

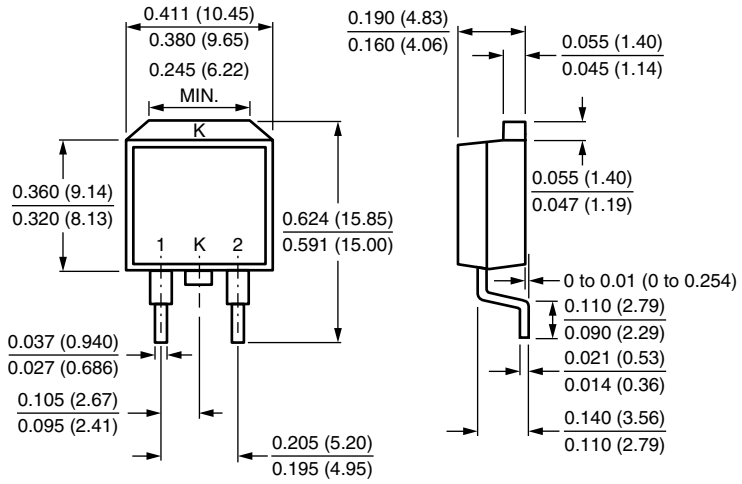


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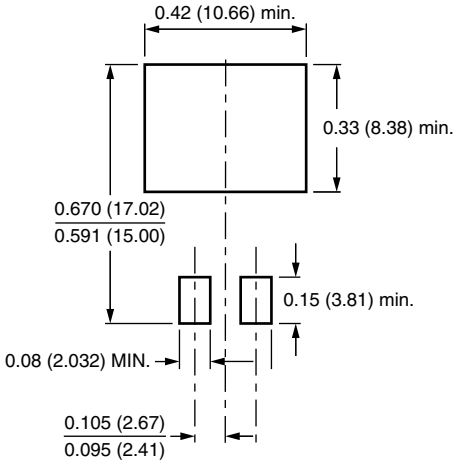
TO-262AA



D²PAK (TO-263AB)



Mounting Pad Layout



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